

B5 MAX Metal Bath Instrument

Software Operation Manual

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1 Overview

1.1 Manual Purpose

The purpose of this manual is to enable users to correctly, safely, and effectively use the Metal Bath Heating and Cooling Control System, achieving precise temperature control to meet laboratory and industrial application requirements.

1.2 Icon Definitions

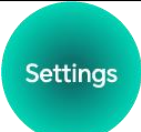
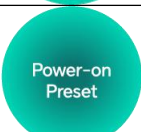
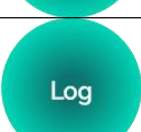
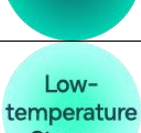
The following table defines the icons used in the system interface:

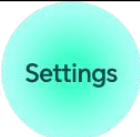
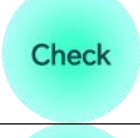
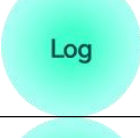
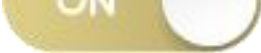
Icon	Icon Meaning
 Manual Setup	Manual Setup (Highlighted)
 Program Control	Program Control (Highlighted)
 Program Edit	Program Edit (Highlighted)
 Operation Curve	Operation Curve (Highlighted)
 System Settings	System Settings (Highlighted)
 Manual Setup	Manual Setup (Normal)
 Program Control	Program Control (Normal)
 Program Edit	Program Edit (Normal)
 Operation Curve	Operation Curve (Normal)
 System Settings	System Settings (Normal)
 Manual Setup	Manual Setup (Disabled)

 Program Control	Program Control (Disabled)
 Program Edit	Program Edit (Disabled)
 Operation Curve	Operation Curve (Disabled)
 System Settings	System Settings (Disabled)
 Prompt	Warning message - Please read before operation
 Start	Start(Highlighted)
 Pause	Pause(Highlighted)
 Stop	Stop(Highlighted)
 Start	Start(Normal)
 Pause	Pause(Normal)
 Stop	Stop(Normal)
 Start	Start(Disabled)
 Pause	Pause(Disabled)
 Stop	Stop(Disabled)
 4°C	Common Temp: 4 ° C (Normal)

37°C	Common Temp: 37 ° C (Normal)
65°C	Common Temp: 65 ° C (Normal)
85°C	Common Temp: 85 ° C (Normal)
Return to Room Temp	Restore Room Temp (Normal)
4°C	Common Temp: 4 ° C (Highlighted)
37°C	Common Temp: 37 ° C (Highlighted)
65°C	Common Temp: 65 ° C (Highlighted)
85°C	Common Temp: 85 ° C (Highlighted)
Return to Room Temp	Restore Room Temp (Highlighted)
⚡ Rate Mode:	Rate Mode: Basic, High, Extreme
🗑 Module Model:	Module Model
Basic	Basic (Normal)
High	High(Normal)
Extreme	Extreme(Normal)
Basic	Basic(Highlighted)

High	High(Highlighted)
Extreme	Extreme(Highlighted)
Confirm	Confirm (Normal)
View	View (Normal)
Delete	Delete (Normal)
View	View (Highlighted)
Delete	Delete (Highlighted)
◀ Previous Page	Previous Page (Normal)
Next Page ▶	Next Page (Normal)
Back	Back (Normal)
Rename	Rename (Normal)
Edit	Edit (Normal)
Save	Save(Normal)
+	Add Program (Normal)

	Delete Program (Normal)
	Low-temperature Storage (Normal)
	Settings (Normal)
	Power-on Preset (Normal)
	Export (Normal)
	Check(Normal)
	Log (Normal)
	Chinese/English (Normal)
	Heated Lid Mode(Normal)
	Alarm Information (Normal)
	Calibration (Normal)
	Low-temperature Storage (Highlighted)

	Settings (Highlighted)
	Power-on Preset (Highlighted)
	Export (Highlighted)
	Check (Highlighted)
	Log (Highlighted)
	Chinese/English (Highlighted)
	Heated Lid Mode(Highlighted)
	Alarm Information (Highlighted)
	Calibration (Highlighted)
	Cold Storage, Startup Preset, Hot Lid, Touch Beep Status ON (Highlighted)
	Cold Storage, Startup Preset, Hot Lid, Touch Beep Status OFF (Disabled)
	Log Details (Disabled)
	Log Details (Highlighted)

		Clear (Normal)
		Low-temperature Storage (Normal)
		Power-on Preset(Normal)
		Heated Lid Mode (Normal)
		Low-temperature Storage (Highlighted)
		Power-on Preset(Highlighted)
		Heated Lid Mode (Highlighted)

2 System Startup

After the device is powered on, the screen first displays the boot screen (approximately 3 seconds), then automatically enters the power-on self-check process. The system will detect whether the temperature sensor is working properly within 10 seconds. After the self-check passes, it automatically enters the manual settings page.



Figure 2-1 Power-on self-check Page

After the self-check passes, as shown in the figure below, the main manual settings page is displayed, and operation can begin.

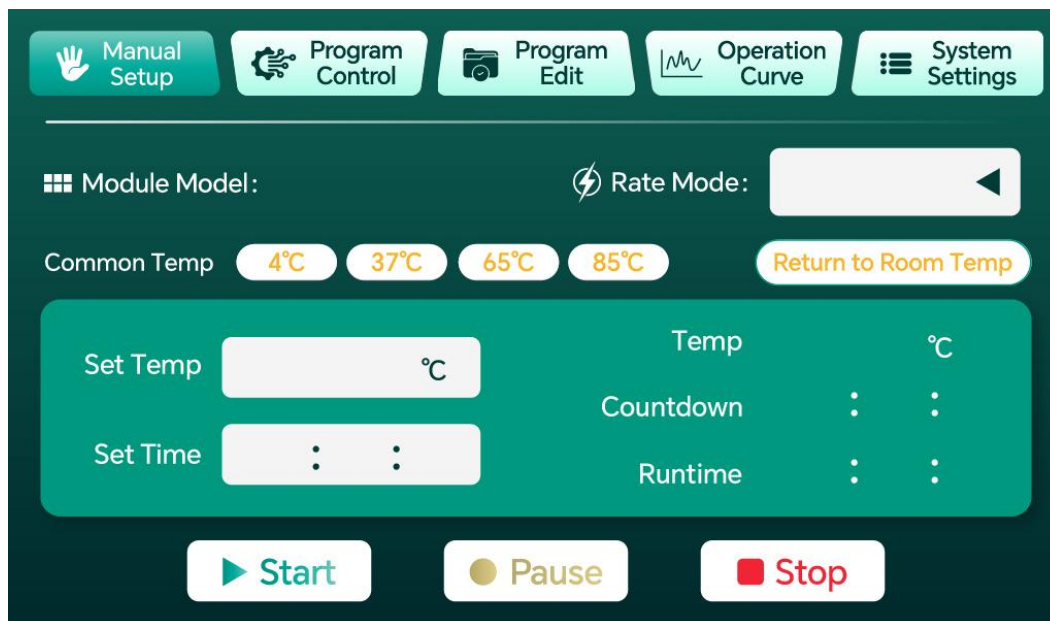


Figure 2-2 Manual Setting Main Page

3 Manual Setting

3.1 Manual Settings Page Overview

The manual settings page is the main operating interface of the device. Users can complete temperature setting, time setting, rate mode selection, and device start/stop control on this page.

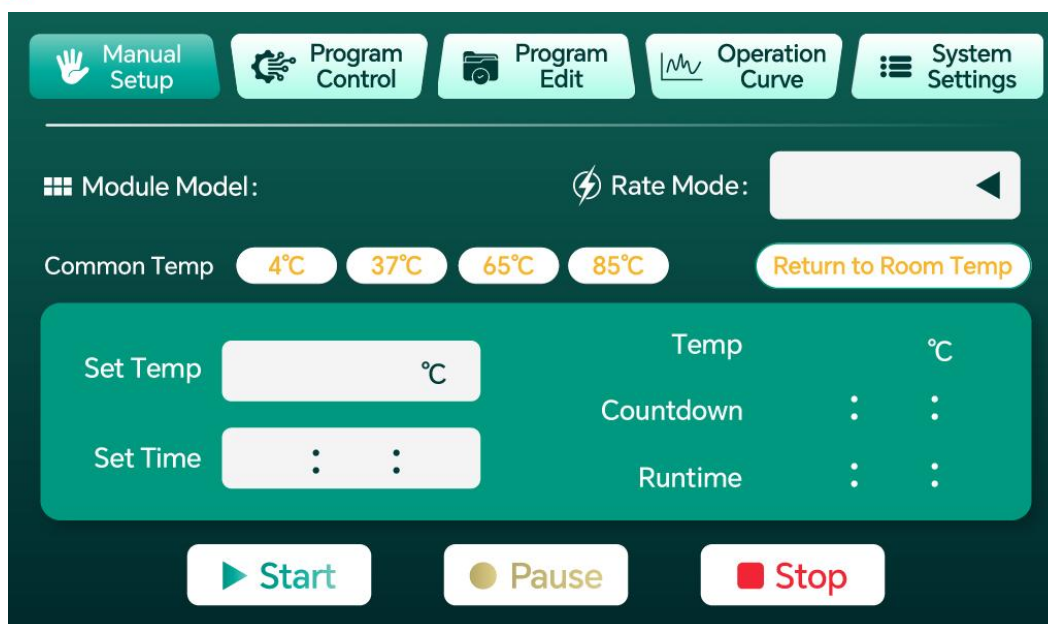


Figure 3-1 Manual Setting Page

- **Module Model:** Displays the currently detected module model (automatically identified via hardware ID pins, supports up to 16 module types).
- **Rate Mode:** Select from three heating/cooling rate levels: Basic Mode, High-Speed Mode, and Extreme Mode.
- **Common Temperatures:** Provides four shortcut buttons for commonly used temperatures (4°C, 37°C, 65°C, 85°C). Click to quickly set the target temperature.
- **Return to Room Temperature:** One-click return of the sample temperature to 23°C. When the temperature reaches 23±1°C, it automatically stops, the button pops up, and the buzzer sounds once.
- **Set Temperature:** Click the input box to set the target temperature. Range: 4~100°C, accuracy: 0.1°C.
- **Set Time:** Set the hold time (H:MM:SS), maximum 99:59:59.
- **Real-time Temperature:** Displays the current real-time temperature of the sample area.
- **Hold Countdown:** Automatically starts countdown after the temperature reaches the set value.
- **Run Time:** Displays the cumulative running time of the device.
- **Run/Pause/Stop:** Three control buttons for starting, pausing, and stopping the device.

Note: Both the temperature and time settings have power-off memory function. After a power failure, the device will restore the previous settings upon restart.

3.2 Temperature Setting

Click the "Set Temperature" input box to bring up a numeric keypad. Enter the target temperature value and click "Confirm." The temperature range is 4~100°C. An out-of-range value will trigger a prompt message.

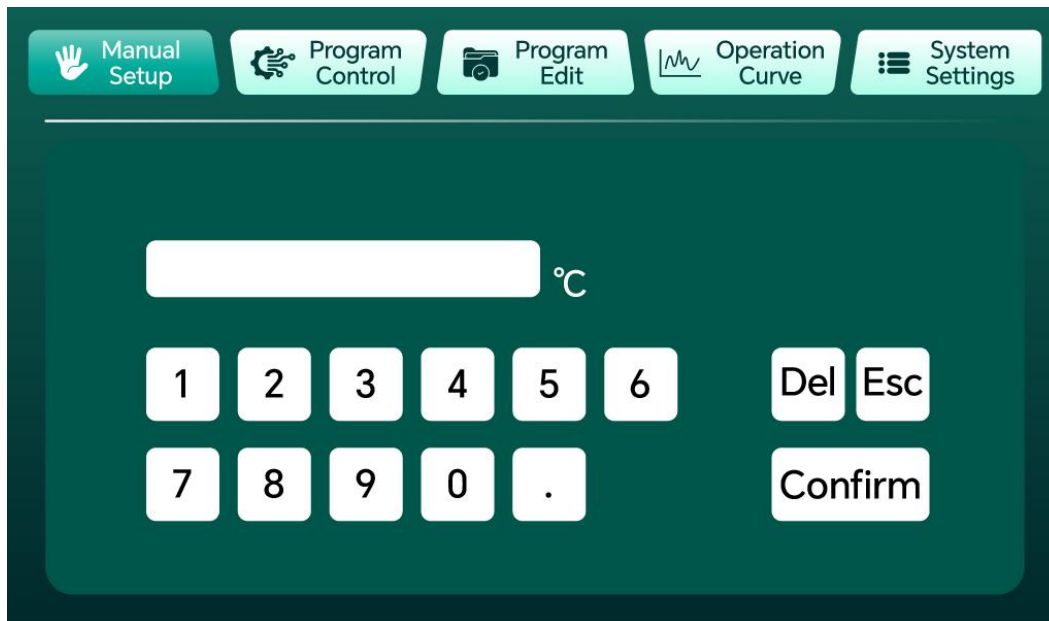


Figure 3-2 Temperature Input Keypad

3.3 Time Setting

Click the "Set Time" input box to bring up the time setting interface. Set hours, minutes, and seconds separately. Maximum settable time is 99:59:59. When the set time is 0, the device will run continuously (no countdown).

3.4 Rate Mode

Tap the "Rate Mode" dropdown to select from the following three heating/cooling rates:

Mode	Heating Rate	Cooling Rate	Recommended Use
Basic Mode	Approx.2°C/min	Approx.0.5°C/min	For experiments not requiring rapid temperature change
High Speed Mode	Approx.5°C/min	Approx.1°C/min	For experiments requiring fast target temperature attainment
Extreme Mode	Unlimited	Unlimited	Reach target temperature at maximum speed

3.5 Common Temperature Shortcuts

The manual settings page provides 4 shortcut buttons for commonly used temperatures: 4°C, 37°C, 65°C, and 85°C. Click the corresponding button, and the target temperature will be automatically set to that value without manual input. When the set temperature is not one of the common temperatures, the common temperature buttons pop up.

3.6 Return to Room Temperature

Click the "Return to Room Temperature" button, and the device will automatically control the temperature to 23°C. When the temperature reaches 23±1°C, it automatically stops, the button pops up, and the buzzer sounds once. This function is suitable for safely removing samples after an experiment.

During the "Return to Room Temperature" process, click "Stop" to cancel the function. When stopped, the fan will continue working for 10 seconds before stopping.

4 Program Control

4.1 Program Control Page Overview

The program control page is used to execute preset multi-step temperature control programs for automated program operation.

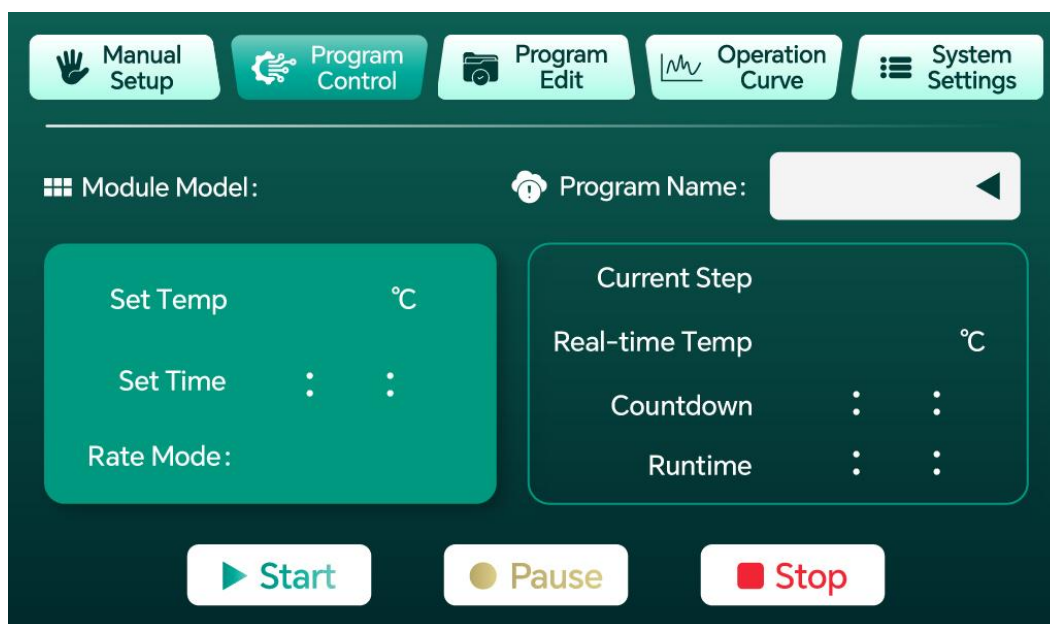


Figure 4-1 Program Control Page

- Set Temperature: Displays the target temperature of the current step (not directly editable; set from the Program Editing page).
- Set Time: Displays the hold time of the current step (H:MM:SS).
- Rate Mode: Displays the heating/cooling rate mode for the current step.
- Program Name: Dropdown menu in the upper right to select the program to run (also selectable from the Program Editing page).
- Current Step: Displays the step number currently being executed.
- Real-time Temperature: Displays the current real-time temperature of the sample area.
- Hold Countdown: Countdown after reaching the temperature of the current step.
- Run Time: Displays the cumulative running time in program mode.
- Run/Pause/Stop: Controls the start and stop of the program.

Note: In program mode, when the current step countdown ends, it will automatically switch to the next step. After all steps are completed, the device will automatically stop and emit a buzzer alert.

4.2 Program Edit

4.2.1 Program Library Page

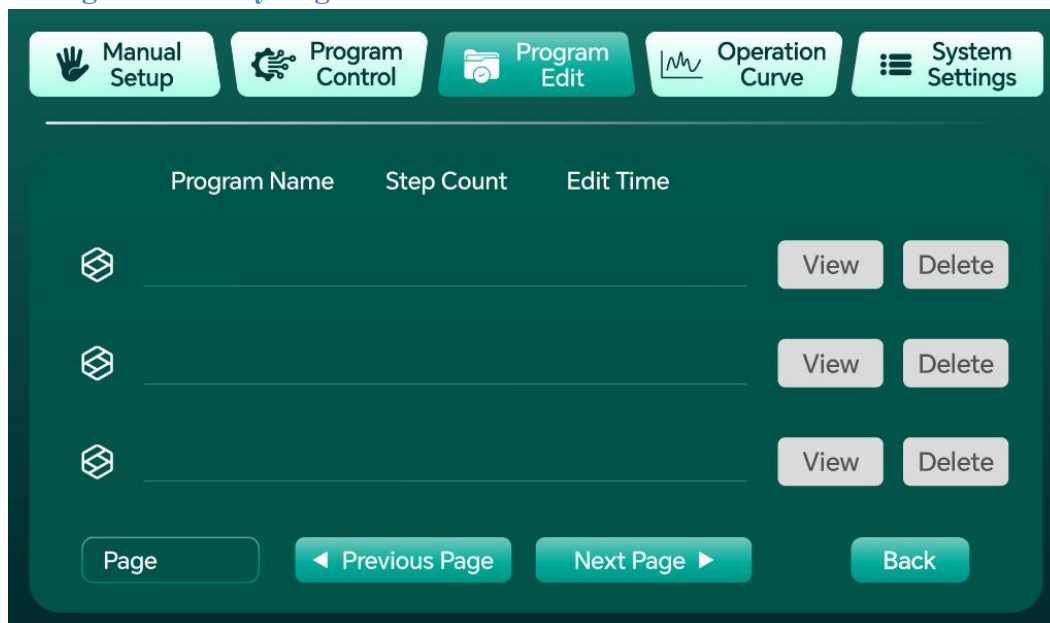


Figure 4-2 Program Edit Page (Program Library)

On the Program Editing page, you can select a program to run and view detailed program steps. Maximum of 5 programs, each containing up to 10 steps. The set programs have power-off memory function. After a power failure, the device will restore the previous settings upon restart.

- Click the "View" button to view the detailed content of the current program.
- Click the "Delete" button to delete the corresponding program.

4.2.2 Program Viewing

After clicking the "View" button, you enter the Program Viewing page, which displays all step details of the program (target temperature, hold time, rate mode). Click the "Program Edit" button to jump to the Program Editing page to edit the current program. The program name can be changed via the dropdown menu.

The screenshot shows the 'Program Edit- ViewPage (Program Library)' interface. At the top, there is a navigation bar with five buttons: 'Manual Setup' (hand icon), 'Program Control' (gear icon), 'Program Edit' (folder icon), 'Operation Curve' (line graph icon), and 'System Settings' (three horizontal lines icon). Below the navigation bar, the main content area has a dark green background. It starts with a 'Program Name:' label, a text input field, a 'Rename' button, and a 'Back' button. Below this, there is a table with four columns: 'Step', 'Set Temp (°C)', 'Rate Mode', and 'Set Time'. The table contains three rows of data, each with a light blue rounded rectangular cell. The first two columns of each row are empty, and the last two columns contain colons. At the bottom of the interface, there is a 'Page' button, a 'Previous Page' button with a left arrow, a 'Next Page' button with a right arrow, and an 'Edit' button.

Step	Set Temp (°C)	Rate Mode	Set Time
			:
			:
			:

Figure 4-3 Program Edit- ViewPage (Program Library)

4.2.3 Program Edit Page

Program Name: Back

Step	Set Temp (°C)	Set Time	Rate Mode
+			
+			
+			

Page ◀ Previous Page Next Page ▶ Save

Figure 4-4 Program Edit- EditPage (Program Library)

- PageLeftside has 3 plus (+) buttons. When a button is pressed, the corresponding row text fields become editable. When the button is released, the corresponding row content is deleted and subsequent steps shift forward automatically. (10 steps max; must be added in sequence; the next button is disabled until the previous one is pressed.)
- Steps are paginated via "Previous Page" / "Next Page". Each program can save a maximum of 10 steps.
- After editing, tap "Save" to save all step settings for the current program.
- If time or temperature is 0, saving is not allowed and a popup warning appears. A popup also confirms successful save. Temperature must be within 4~100°C and set time cannot be 0, otherwise a warning will pop up.

Note: Tap "Cancel" or "Back" to return to the Program Library page.

5 Operation Curve



Figure 5-1 Operation Curve Page

The running curve page is used to display temperature change trends in real time, helping users intuitively understand the temperature control process.

- The vertical axis is temperature ($^{\circ}\text{C}$), and the horizontal axis is time (min).
- Common temperature reference lines are marked on the right side: 4°C , 37°C , 65°C , 85°C , 100°C .
- Records up to 12,800 data points. Each page displays approximately 1 hour of data (720 points).
- Navigate through Page X/Previous Page/Next Page buttons to view historical temperature trends. Supports quick jump navigation.
- The curve updates in real-time during operation. Historical data can be reviewed after stopping.

Note: Page navigation (Page X/Previous Page/Next Page) is only available after the current page has finished rendering.

6 System Settings

6.1 System Settings Page Overview

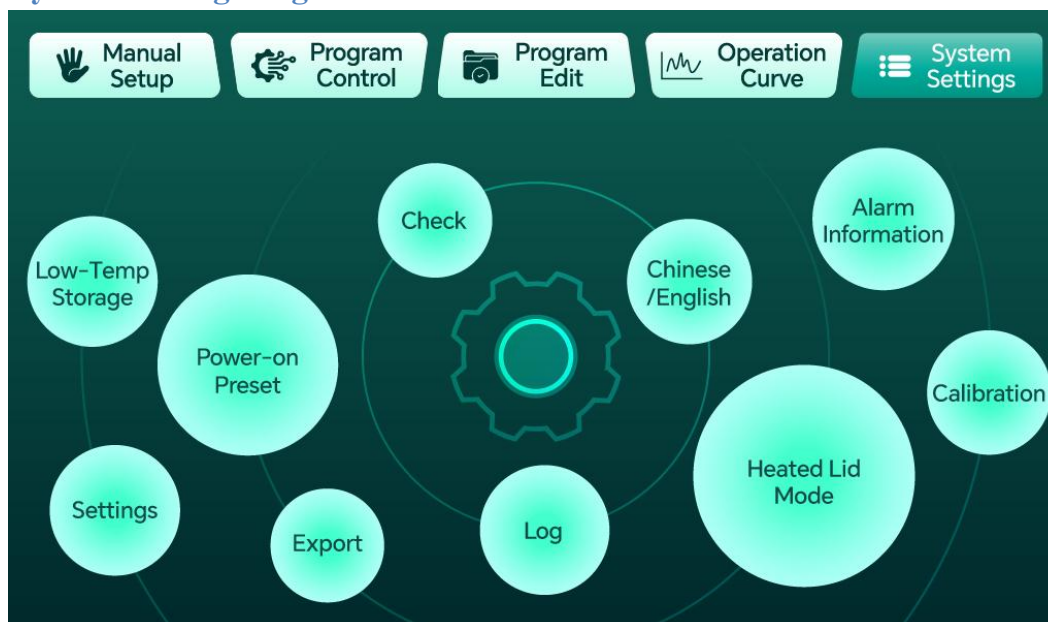


Figure 6-1 System Settings Page

- When clicking System Settings during manual settings or program operation, all icons except Settings and Calibration are disabled.
- After entering the Settings or Calibration menu from System Settings, the Manual Settings and Program Editing menus are grayed out.

The System Settings page uses a circular icon layout and includes the following sub-function modules:

No.	Function Module	Description
1	Low-temperature Storage	Automatically enters low-temperature storage mode after operation ends
2	Power-on Preset	Automatically pre-heats/pre-cools to the set temperature after power-on
3	Check	Re-execute the power-on self-check process
4	Chinese/English	Interface language switch (Chinese/English) - not yet enabled in current version
5	Heated Lid Mode	Independent control of hot lid temperature
6	Alarm Information	View historical alarm

		records
7	Calibration	Temperature compensation calibration
8	Settings	RTC time setting, restore default settings
9	Export	USB/SD card data export
10	Log	View running log records

Note: When entering the Calibration page from System Settings, a password is required.

6.2 Low-temperature Storage

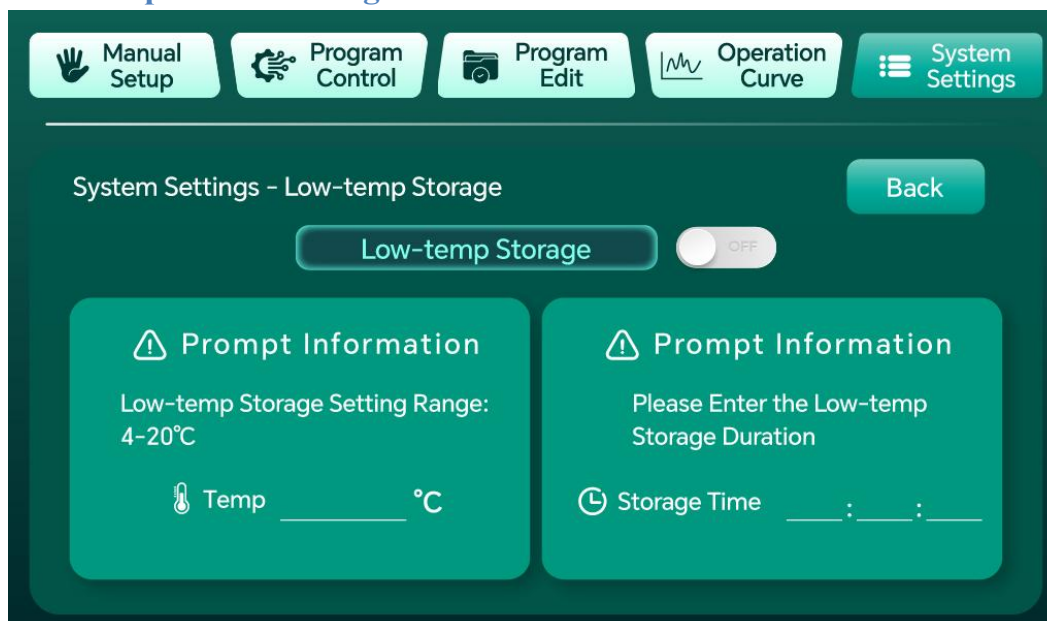


Figure 6-2 Cold Storage Settings Page

Low-temperature storage refers to the function where the instrument automatically lowers the module temperature to a specific temperature (usually 4°C) and maintains it after completing the main experiment program, preventing sample degradation.

- Low Temp Storage Switch: ON/OFF toggle. When enabled, the device will automatically enter low-temperature storage mode after operation ends.
- Low Temp Storage Temperature: Setting range 4~20°C.
- Low Temp Storage Time: Set the storage duration (H:MM:SS). Automatically exits low-temperature storage mode when time expires.

Operating Logic:

- Automatic Transition: When the user's main program (whether in Manual Hold or Automatic Program mode) finishes, if the low-temperature storage function is enabled, the system will seamlessly transition to the low-temperature storage phase.

- **Timed Mode:** If the user sets a low-temperature storage time (e.g., 60 minutes), the instrument will maintain the low temperature for that duration, then automatically stop and shut off temperature control (standby).
- **Continuous Mode:** If no time is set (or time is set to 0/infinite), the instrument will continuously maintain the low-temperature state until the user manually stops it.

6.3 Power-on Preset

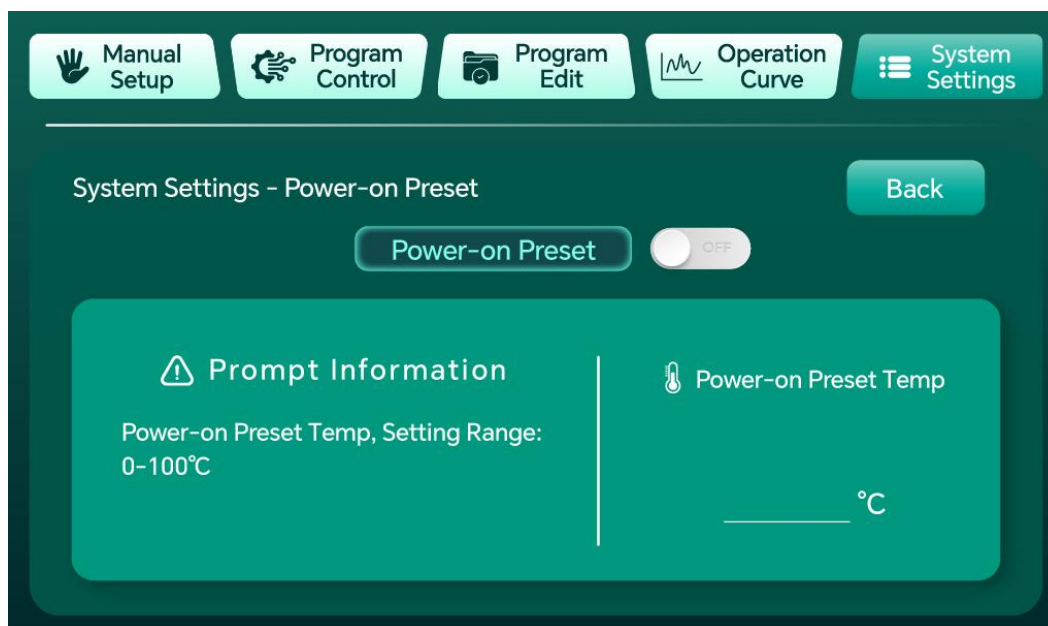


Figure 6-3 Startup Preset Settings Page

Power-on preset refers to the function where the instrument automatically executes a pre-set constant temperature program after power-on, allowing the instrument to quickly reach a common standby temperature (such as 25°C or 37°C), reducing the user's waiting time before experiments.

- **Power-on Preset Switch:** ON/OFF toggle.
- **Power-on Preset Temperature:** Setting range 4~100°C. When enabled, the device automatically starts temperature control to the preset temperature upon power-up.

Operating Logic:

- **Auto-start on Power-up:** After the instrument is powered on and passes the self-check, if the power-on preset switch is detected as enabled, it will directly enter the preset constant temperature running state without user intervention.
- **Priority and Interruption:** This mode has lower priority than the user's main program. During power-on preset operation, once the user starts a new main program (e.g.,

Program A), the power-on preset will be immediately terminated, and the instrument will prioritize executing the user's specified main program.

6.4 Heated Lid Mode

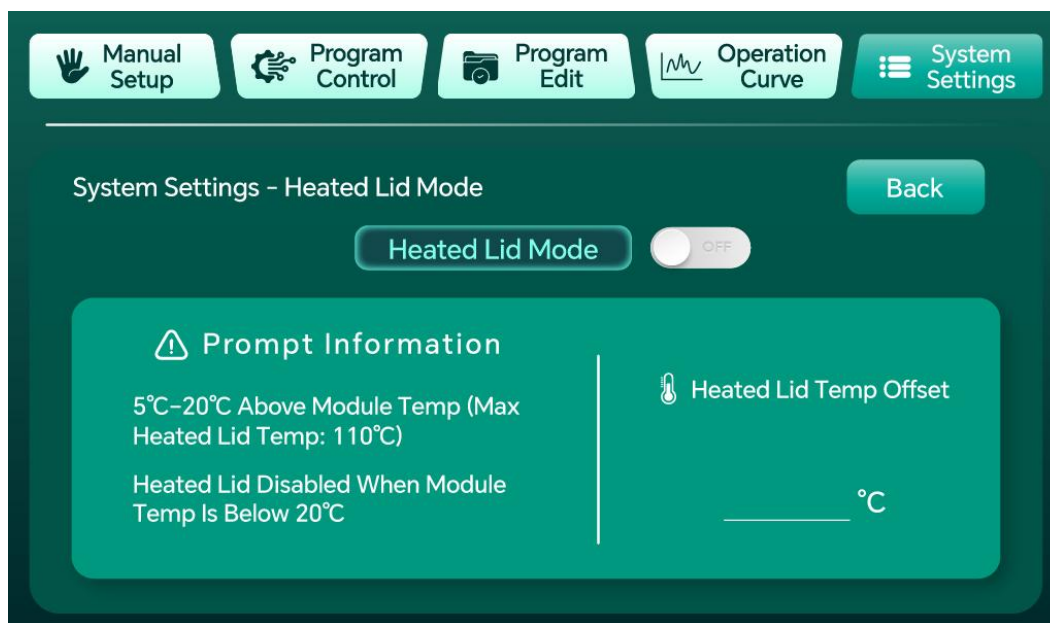


Figure 6-4 Hot Lid Mode Settings Page

The hot lid mode is used to independently control the temperature of the hot lid, preventing condensation on the inner wall of the sample lid during experiments such as PCR.

- Hot Lid Mode Switch: ON/OFF toggle.
- Hot Lid Temperature Differential: Set the temperature difference between the hot lid and the module temperature (5~20°C), maximum not exceeding 110°C.
- Note: When the module temperature is below 20°C, the hot lid function is disabled.

Heated Lid Mode Operation Logic:

- Manual Mode: Turn on the hot lid mode switch and start the run program. The hot lid icon changes from white to yellow, indicating the hot lid function is active. Tap the Stop button to change the icon back from yellow to white.
- Program Control Mode: In Program Control Mode with the hot lid switch on: If target temperature > 20°C, the hot lid activates and the icon changes from white to yellow. If

target temperature < 20°C, the hot lid does not operate and the icon stays white. Press Stop to change the yellow icon back to white.

●If the hot lid is lifted during program operation, the system displays a popup: "Hot lid not detected. Continue running?":

Tap [Yes]:: Disable hot lid function, hide hot lid status icon, program continues (running without hot lid mode).;

Tap [No]:: Disable hot lid function, hide hot lid status icon, terminate the current program.

Note: When hot lid mode is active, Target Temperature = Current Target Temperature - Hot Lid Temperature Differential x 5%.

6.5 Log

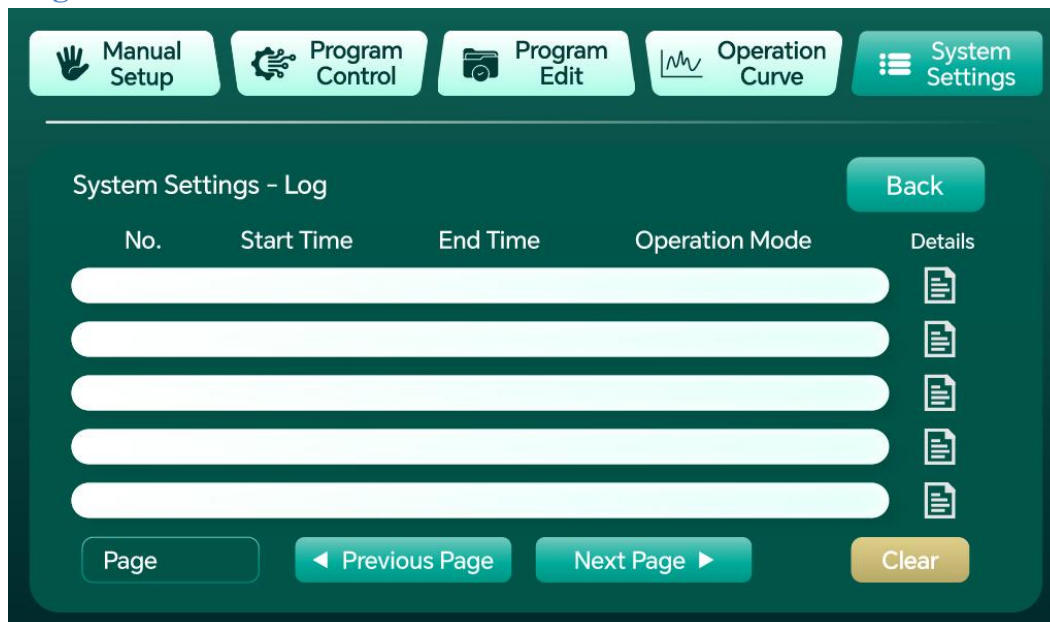


Figure 6-5 Log Page

- Each row displays: No., Start Time (H:M:S), End Time (H:M:S), Run Mode (Manual Mode/Program Mode), Details.
- Click the detail icon to enter the Log Detail page, showing detailed operation information for that log entry.
- Maximum of 100 log entries. Navigate via "Previous Page"/"Next Page" buttons.
- Logs have power-off memory function.
- Click the "Clear" button to clear all log entries. (Note: This process takes some time.)

Note: The Log Detail page is divided into Manual and Program tabs.

6.6 Alarm Information

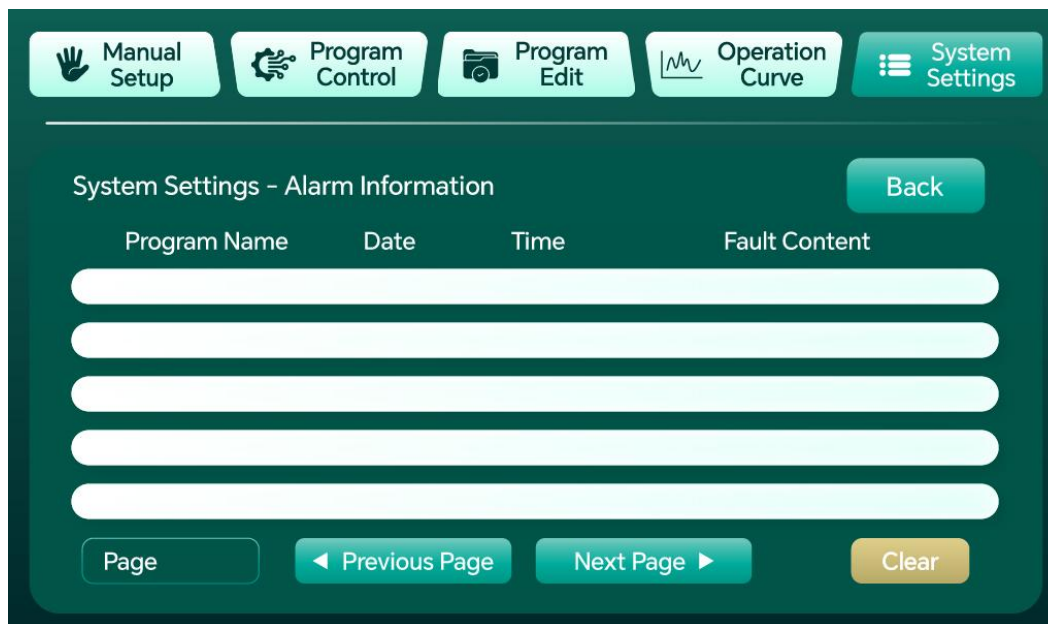


Figure 6-6 Alarm Info Page

- Displays alarm information. Maximum of 100 entries. Navigate via "Previous Page"/"Next Page" buttons.
- Alarm information has power-off memory function.
- Click the "Clear" button to clear all alarm entries. (Note: This process takes some time.)

Alarm types include:

Alarm Type	Description
Temperature Sensor Fault	No valid temperature data detected during power-on self-check
Temperature Anomaly	Temperature remains unchanged for an extended period during operation and has not reached the target temperature
Module Not Properly Installed	No module ID detected
Communication Fault	Serial screen communication malfunction
Memory Fault	FRAM read/write malfunction
Hot Lid Not Detected	Hot lid module not connected

6.7 Self-check

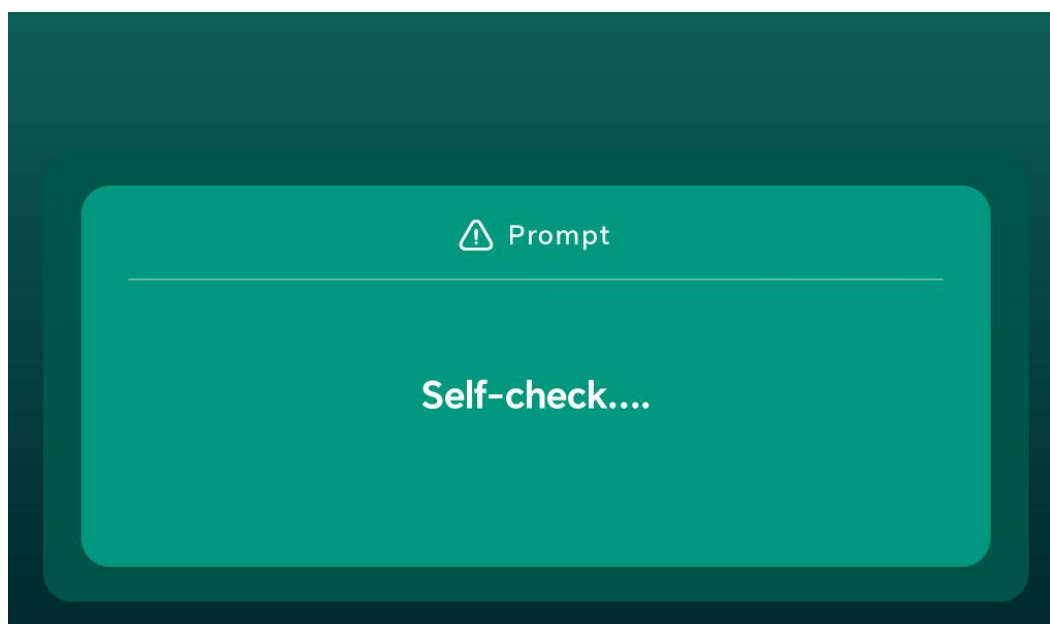


Figure 6-7 Self-Test Page

Click the "Self-check" button, and the device will re-execute the power-on self-check process, checking whether the temperature sensor is working properly (whether valid temperature data is read within 10 seconds). After the self-check passes, it returns to the System Settings page. Fan speed is at 50% during self-check.

6.8 Calibration (Temperature Compensation)

Manual Setup

Program Control

Program Edit

Operation Curve

System Settings

System Settings - Calibration

Next Page

Back

Room Temperature (PT1)		°C	4		4		
Upper Module (PT2)		°C	Large Module	37		37	
Lower Heating Plate (PT3)		°C		65		65	
Heat Sink Fin (PT4)		°C		85		85	
Fan Threshold		°C		100		100	

Small Module

Temperature Compensation (°C)

The calibration function is used to compensate and calibrate temperature measurement values, improving temperature control accuracy. The system supports 4 sets of temperature compensation values, corresponding to 4 temperature zones:

Temperature Zone	Range
Low Temp Zone	0~15°C (including 0°C)
Room Temp Zone	15~55°C (including 15°C and 55°C)
Medium Temp Zone	55~85°C
High Temp Zone 1	85~100°C (including 85°C)
High Temp Zone 2	100~105°C (including 100°C)

Users can adjust the compensation value for the corresponding zone based on the reading of a standard thermometer to make the displayed temperature more accurate.

- 5mlx8, 15mlx8, 50mlx4 are large modules and use large module temperature compensation.
- 0.6mlx24, 1.5mlx24, 2.0mlx24, and well plate modules are small modules and use small module temperature compensation.

6.9 Settings

The Settings page includes the following functions:

- RTC Time Setting: Set the real-time clock of the device (Year/Month/Day/Hour/Minute), providing accurate timestamps for log and alarm records.
- Restore Default Settings: Restore all parameters to factory default values.

6.10 Data Export

- USB Export: Supports exporting running log data to a USB drive (Excel format) via the USB interface, for convenient data analysis on a computer.
- SD Card Auto-save: Supports periodic automatic saving of temperature data to the SD card, once every 5 minutes.

6.11 Product Information

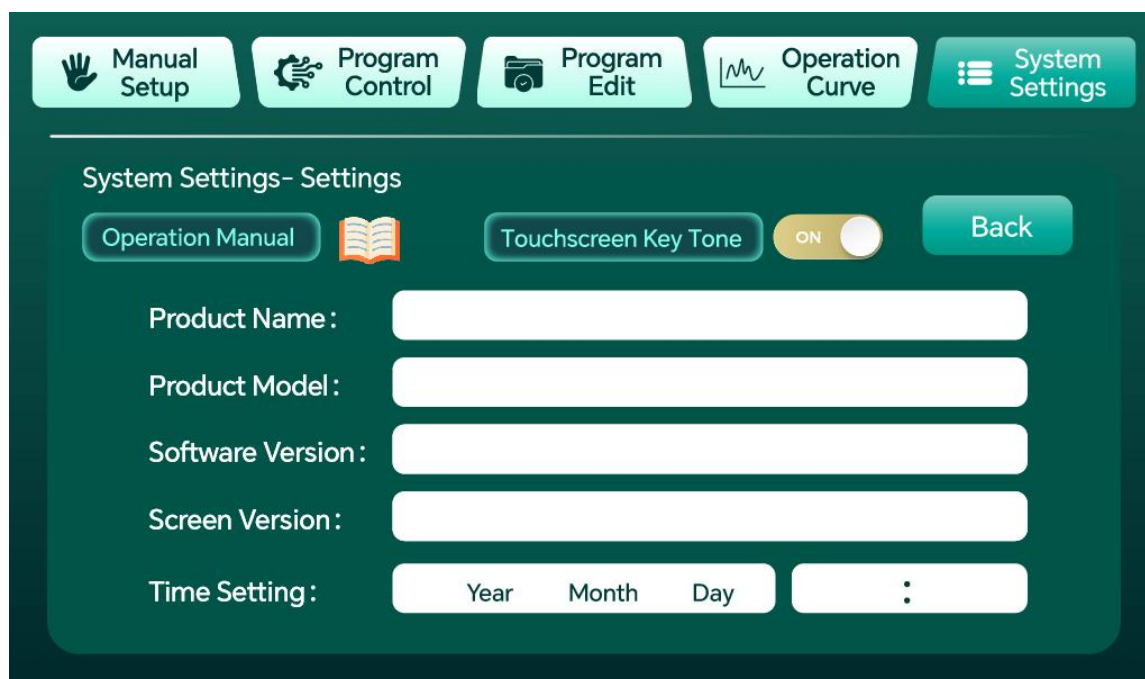


Figure 6-8 Device Info Page

Displays basic product information:

Item	Content
Product Name	B5 MAX Intelligent Metal Bath
Product Model	B5 MAX
Software Version	B5.MAX-V3.3.2
Screen Version	V6.7.1
Time Setting	Set the machine's year, month, day, hour, and minute

7 Operating Instructions

7.1 Running Mode Introduction

The Metal Bath Heating and Cooling Instrument supports two main operating modes:

- (1) Manual Mode: Operation method - Set the target temperature and hold time on the Manual Settings page, then click the "Run" button to start. Stop method - Click the "Stop" button, or the device automatically stops when the hold time expires.
- (2) Program Mode: Operation method - Select or edit a program on the Program Editing page, set the target temperature, hold time, and rate mode for each step, then

return to the Program Control page and click "Run." Stop method - Click the "Stop" button, or the device automatically stops when all steps are completed.

7.2 Manual Mode

7.2.1 Preparation

- Confirm mode: Manual Settings page
- Set the target temperature (4~100°C)
- Set the hold time (optional; set to 0 for continuous operation)
- Select the rate mode (Basic/High/Extreme)

7.2.2 Operation Method

Click "Run" - the device starts heating/cooling to the target temperature. The Run button lights up, and the Stop button turns off.

When the temperature reaches the set value, the hold countdown automatically starts.

When the hold countdown ends, the buzzer beeps three times, and the device automatically stops.

During operation, you can click the "Pause" button at any time to pause, and click "Run" again to resume.

During operation, you can click the "Stop" button at any time to stop.

Notes:

- 1. If a temperature anomaly occurs during operation, the device will automatically stop and display alarm information on the alarm page, while recording the alarm information.
- 2. Power-off Resume Function: If an unexpected power failure occurs during operation, the device will automatically resume operation from the breakpoint when power is restored.
- 3. If the module is not properly installed, the module model displays "Module Not Detected," and Run/Pause/Stop buttons are all grayed out (both manual and program modes).
- 4. During operation, the temperature and time input boxes cannot be edited, and the rate mode cannot be modified.

7.3 Program Mode

7.3.1 Preparation

- Confirm mode: Program Control page
- Select the program to run on the Program Editing page
- Confirm that the step parameters (target temperature, hold time, rate mode) are correctly set

7.3.2 Operation Method

Click "Run" - the device starts executing the first step, heating/cooling to the target temperature of that step.

When the temperature of the current step is reached, the hold countdown starts. When the countdown ends, the device automatically switches to the next step, and the buzzer beeps once to indicate step switching.

The current step number, target temperature, and hold countdown are updated in real time.

After all steps are completed, the buzzer beeps three times, and the device automatically stops.

Notes:

1. During operation, click the "Stop" button at any time to stop.
2. When restarted after stopping, the current step, target temperature, and hold time will be restored to those of Step 1 of the program.
3. If a temperature anomaly occurs during operation, the device will automatically stop and display alarm information on the alarm page, while recording the alarm information.
4. Power-off Resume Function: If an unexpected power failure occurs during operation, the device will automatically resume from the breakpoint (including the current step and remaining time) when power is restored.
5. The program name cannot be changed during program operation.
6. The lower heating plate is not allowed to exceed 108°C (physical limit; output may stop if exceeded).

8 Technical Specifications

Category	Parameter
Main Control Chip	STM32F407VETx (168MHz Cortex-M4F)
Operating System	FreeRTOS 10.x Real-Time OS
Display Screen	7-inch DGUS Serial Touch Screen
Temperature Sensor	PT1000 Platinum RTD (4 channels)
ADC Accuracy	AD7124 24-bit High-Precision ADC

Storage	FRAM FM24V05 64KB (non-volatile)
Control Algorithm	Dual-loop cascaded PID (incremental + positional)
Communication Protocol	Modbus, DGUS serial display protocol
Temperature Range	0°C ~ 105°C
Heating Performance	Three heating modes: Full Power: >5°C/min High: 5°C/min Medium: 2°C/min Heating Time: ≤15 min (room temp 25°C → 100°C)
Cooling Performance	Three cooling modes: Full Power: ≥2°C/min High: ≥1°C/min Medium: ≥0.5°C/min Cooling Time: ≤15 min (room temp 25°C → 4°C)
Compatible Modules	Supports various tube sizes (0.6, 1.5, 2, 5, 15, 50 mL) Multi-well plates (cell culture plates, ELISA plates, full-skirted microplates) Easy module replacement, no tools required
Compatible Lids	Standard module lid included (precision temperature control) Optional hot lid (prevents low-temperature condensation, 1.5/2mL)
Program Control	Up to 50 programs, each with up to 10 steps
Log Capacity	10 run logs + 10 alarm records
Temperature Curve	Maximum 12,800 data points
Software Scale	Approximately 21,000 lines of C code

Tips

1. Before turning off the instrument, stop the device first and wait for the temperature to return to room temperature before switching off the power.
2. Do not restart the instrument immediately after turning off the power. Improper operation will affect the service life of the instrument.

Scan QR Code to View Instructions

(Scan with WeChat or mobile browser to view instrument operation instructions)

